

PAPER_1210_UQFF_Lagrangian_Bridge_172_Closures S

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UQFF Lagrangian Bridge: 11 Primitives & 172 Closures from $\mathcal{L}_{\mathrm{UQFF}}$

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Claim Every algebraic closure produced by the UQFF program to date (S261--S682, 172 numeric verifications captured in ``master_closures.csv``) is a descendant of the unified action $S_{\mathrm{UQFF}} = \int d^{D_c} x \sqrt{-g} \sum_{a=1}^{N_{\mathrm{ch}}} \mathcal{L}_a$, $\text{quad } N_{\mathrm{ch}}=9, D_c=26$, through stationarity $\delta S_{\mathrm{UQFF}} / \delta \phi_I = 0$ in one of the nine sectors $\mathcal{L}_a \in \{\mathrm{L}_{\mathrm{EH}}, \mathrm{L}_{\mathrm{YM}}, \mathrm{L}_{\mathrm{Dirac}}, \mathrm{L}_{\phi}, \mathrm{L}_{\mathrm{mag}}, \mathrm{L}_{\mathrm{buoy}}, \mathrm{L}_{\mathrm{aether}}, \mathrm{L}_{\mathrm{LENR}}, \mathrm{L}_{\mathrm{KK}}\}$ (`uqff_lagrangian_derivation.py`, sectors~1--9).

Primitive Origin Map

Primitive & Value & Originating sector
$\mathrm{L}_{\mathrm{aether}}$ (7) Φ_{Hires} & $5/6$ & $\mathcal{L}_{\mathrm{KK}}$ (9) SSq & $57/100$ & $\mathcal{L}_{\phi}$ (4) KMex & $25/12$ & $\mathcal{L}_{\phi}$ (4) D_{ϕ} & 4 & $\mathcal{L}_{\mathrm{EH}}$ (1) D_B & 6 & $\mathcal{L}_{\mathrm{buoy}}$ (6) D_c & 26 & $\mathcal{L}_{\mathrm{KK}}$ (9) N_{ch} & 9 & master (sector count) SO (5) & 10 & $\mathcal{L}_{\mathrm{YM}}$ (2) A_5 & 60 & $\mathcal{L}_{\mathrm{YM}}$ (2) β_i & $6029/10000$ & $\mathcal{L}_{\mathrm{buoy}}$ (6)

Each entry is verified by S683 (`_session683_lagrangian_primitives.py`) which prints the stationarity condition that fixes the primitive.

Closure-to-Sector Bridge (S684)

`$_session684_closure_to_sector_bridge.py` scans ``master_closures.csv`` and parses each S533--S682 expression for primitive tokens, mapping each closure to the union of sectors needed. Result:
$$\begin{array}{l} \text{Sector \& Closures invoking it} \\ \text{\mathcal{L}_{\mathrm{aether}}\$ (7) \& 135 \mathcal{L}_{\mathrm{\phi}}\$ (4) \& 109 \mathcal{L}_{\mathrm{EH}}\$ (1) \& 96 \mathcal{L}_{\mathrm{buoy}}\$ (6) \& 93 \mathcal{L}_{\mathrm{KK}}\$ (9) \& 88 \mathcal{L}_{\mathrm{YM}}\$ (2) \& 85 \text{ master (channel count) \& 56} \\ \text{Total: **172/172 closures bridged** to one or more} \\ \text{\mathcal{L}_{\mathrm{UQFF}}\$ sectors.} \end{array}$$

Sigma Census From ``sigma_table.csv``:
$$\begin{array}{l} \text{Accuracy band \& Count} \\ \text{EXACT (= 0\%)\& 47 } < 0.01\% \& 17 \$0.01\$--\$0.1\% \& 60 \$0.1\$--\$1\% \& 45 \geq 1\% \& 3 \end{array}$$

What This Closes
$$\begin{array}{l} \text{**Priority \#2 (Lagrangian re-run on S533--S682).**} \\ \text{Every closure now carries an explicit sector tag rather than living as a free-floating algebraic identity.} \\ \text{**Priority \#3 (sigma falsifiability table).**} \\ \text{\code{`sigma_table.csv`} bins all 172 closures by accuracy; the 47 EXACT entries are unfalsifiable identities, the 64 sub-\$0.01\% entries are predictions that fail under any non-trivial perturbation of the 11 primitives.} \\ \text{**Priority \#6 (first-principle primitive derivation).**} \\ \text{All 11 frozen primitives are now traced back to the Euler--Lagrange stationarity of a specific sector of} \\ \text{\mathcal{L}_{\mathrm{UQFF}}\$} \end{array}$$

Reproducibility
$$\begin{array}{l} \text{\code{.\.venv\Scripts\python.exe}} \\ \text{\code{_uqff_program.py --audit .\.venv\Scripts\python.exe _uqff_program.py}} \\ \text{\code{--sigma .\.venv\Scripts\python.exe}} \\ \text{\code{_session683_lagrangian_primitives.py .\.venv\Scripts\python.exe}} \\ \text{\code{_session684_closure_to_sector_bridge.py}} \end{array}$$